Appendix I
Training and Implementation Details

Table V lists the training hyperparameters for all pipeline components. The world model consists of a convolutional VAE operating on 64×64 images and a ConvLSTM predictor that autoregressively rolls out latent states given a context of m=100 frames. Both are trained with AdamW and cosine-annealed learning rates. The CNN safety classifier is trained on latent representations with a binary cross-entropy objective.

For post-hoc calibration, per-horizon temperatures T_K are obtained by minimizing negative log-likelihood (NLL) via bounded scalar optimization on the calibration split. The anomaly-conditioned calibrator weights (w_ρ, w_δ) are fit on the pooled calibration set $\mathcal{D}_{\text{in}}^{\text{cal}} \cup \mathcal{D}_{\text{aug}}$ (5-fold CV for robustness), with all real OOD data held out for evaluation. The Mahalanobis distance for the dynamics score uses a covariance regularization of λ=10^{-6} with StandardScaler pre-processing.

TABLE V: Training hyperparameters for each pipeline component.

Component	Hyperparameters
VAE (64×64)	AdamW, lr $= 10^{-3}$, weight decay 10^{-5}, batch 32, 50 epochs, latent tensor $4\times4\times64$ (64 channels)
LSTM predictor	AdamW, lr $= 10^{-4}$, weight decay 10^{-4}, batch 8, 60 epochs, hidden 128, context m=100, cosine annealing ($\eta_{\min}$=10^{-6})
CNN classifier	AdamW, lr $= 10^{-3}$, weight decay 10^{-5}, batch 128, 50 epochs
Temperature scaling	Per-horizon T_K via bounded scalar minimization of NLL
Anomaly calibrator	L-BFGS-B, $w_\rho, w_\delta \geq 0$, 5-fold CV on $\mathcal{D}_{\text{in}}^{\text{cal}} \cup \mathcal{D}_{\text{aug}}$
Mahalanobis δ	$\lambda = 10^{-6}$, StandardScaler pre-processing

Appendix II
Evaluation Metrics

Evaluation Metrics. We evaluate the calibrator's performance using two complementary metrics: calibration quality (ECE) and discriminative ability (AUROC).

- *Expected Calibration Error (ECE) [1].* We report ECE with B bins:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|\mathcal{I}_b|}{N} \left|\text{acc}(\mathcal{I}_b) - \text{conf}(\mathcal{I}_b)\right|, \tag{8}$$

where $\text{acc}(\mathcal{I}_b)$ and $\text{conf}(\mathcal{I}_b)$ denote the empirical accuracy and mean predicted confidence in bin $\mathcal{I}_b$. Lower ECE indicates better calibration.

- *Area Under the Receiver Operating Characteristic Curve (AUROC)* [9]. We report AUROC for safety confidence scores and OOD detection scores:

$$\text{AUROC} = \int_0^1 \text{TPR}\big(\text{FPR}^{-1}(u)\big)\, du, \tag{9}$$

where TPR is the true positive rate, FPR is the false positive rate, and u is the threshold. A value of 1.0 means perfect discrimination, and 0.5 means coin-flip performance.

Appendix III
Dynamics Feature Vector

Table VI enumerates the 12 features comprising the dynamics feature vector $\mathbf{f}_i^K$, computed over a context window of m_w=100 steps. We define a *freeze* as $|\Delta u_t| < 0.02$ and *active steering* as $|u_t| > 0.15$. Features marked † depend on the prediction horizon K and are recomputed from the MC Dropout sequence for each K. All features are *causal*: they are computable at time i using only the observed context $y_{i-m+1:i}$ and the LSTM forward pass, without requiring future ground-truth observations.

TABLE VI: The 12-dimensional dynamics feature vector $\mathbf{f}_i^K$.

Group	Dim	Features
MC Dropout	3	Rollout-mean†, rollout-max†, volatility† of per-step latent std
Action	6	Max freeze run, cond. freeze run & frac., reversal rate, jerk, mean steering
Temporal	3	Steering–latent xcorr peak, HF spectral ratio, reaction delay

Appendix IV
Safety Classifier Performance

Table II reports the base CNN predictor's AUROC under nominal and anomalous conditions at four prediction horizons. Under nominal driving, the classifier maintains strong discriminative performance (mean AUROC 0.944), confirming that the safety labels are learnable from latent representations. However, all four anomaly types cause substantial degradation, particularly at longer horizons: Dark degrades most severely ($0.986 \rightarrow 0.547$ from K=5 to K=50), consistent with the progressive divergence of VAE reconstructions under low illumination. Dynamics anomalies (Bias, Latency) show a flatter but still significant decline, since the visual input remains plausible while the action–state coupling deteriorates. These trends motivate the need for anomaly-informed calibration: the raw classifier confidences become increasingly unreliable precisely where calibration matters most.

Appendix V
Alternative Calibration Metrics

To verify that the calibration rankings reported in Section IV-B are not artifacts of ECE's fixed-width binning, we re-evaluate all methods under two complementary metrics proposed by Nixon et al. [34]: Adaptive Calibration Error (ACE), which partitions predictions into equal-mass bins to eliminate sensitivity to bin width, and Static Calibration Error (SCE), which independently assesses calibration for both the positive and negative class.

Fig. 5 plots both metrics as a function of prediction horizon K for all four OOD protocols. The relative ranking of methods is fully preserved: our method (TTA+ρ+δ) achieves the lowest error across all conditions, DAC [19] ranks second, and standard post-hoc calibrators (Temperature Scaling, Isotonic Regression, Histogram Binning) cluster at higher error levels. This three-metric consistency—ECE, ACE, and SCE—strengthens the conclusion that anomaly-informed temperature scaling provides genuine calibration improvement rather than exploiting a particular binning strategy.

APPENDIX VI
COMPUTATIONAL COST

Table VII reports per-sample inference time for each pipeline component. All models use FP32 precision; batch size is 1 unless noted.

TABLE VII: Per-sample inference time breakdown. MC Dropout samples are batched (B=3) for GPU parallelism. Model sizes: VAE 5.5M, LSTM 2.4M, CNN 2.6M parameters.

Stage	**Component**	**Time (ms)**
Perception	VAE encode (m=100 frames)	2.7
	VAE decode (1 frame)	2.2
Dynamics	LSTM deterministic (150 steps)	173.7
	MC Dropout (B=3, 150 steps)	279.4
Safety pred.	CNN inference (1 image)	1.5
	TTA (1+8 augmented views)	14.4
Anomaly scores	ρ: VAE reconstruction (100 fr)	52.4
	δ: Mahalanobis (12-d)	<0.01
Calibration	Temperature Scaling	<0.01
	Ours (anomaly-conditioned TS)	<0.01
	DAC (KNN, k=10)	19.5
Pipeline totals		
Base (CNN + Temp Scaling)		180.0
+TTA		192.9
Full (ours: +TTA +MC +ρ +δ)		524.7

The dominant cost is the sequential ConvLSTM rollout: the deterministic pass (100-step priming + 50-step prediction) takes 173.7 ms, and the batched MC Dropout pass adds 279.4 ms. The calibration step itself is negligible (<0.01 ms), confirming that the overhead of anomaly-informed calibration over standard temperature scaling is due entirely to the anomaly score extraction, not the calibration function. Compared to the Base pipeline, our full method adds 344.7 ms (+1.9×), of which 80% is MC Dropout.

a) Online deployment.: The times above reflect *cold-start* evaluation, where each sample is processed independently from a full context window. In a streaming setting, three properties reduce the effective per-frame cost substantially:

1) **Incremental hidden state.** The LSTM maintains a running hidden state across frames. Each new observation requires only a single forward step (≈ 1.2 ms) rather than re-priming over the entire context.
2) **Asynchronous anomaly scoring.** The anomaly scores ρ_i and δ_i characterize sustained distributional shift and need not be recomputed every frame. Updating them every N frames (e.g., N=10 at 10 Hz) amortizes the MC Dropout and VAE reconstruction cost to ≈ 33 ms/frame, running on a separate CUDA stream without blocking the control path.
3) **Decoupled control and monitoring.** The safety prediction (CNN + TTA) runs on the critical control path at ≈ 15 ms/frame (≈ 67 Hz). Anomaly scoring and calibration operate asynchronously as a monitoring layer; the latest available score δ_i is used.

Under this design, the real-time bottleneck is the CNN + TTA pass (≈ 15 ms), which comfortably meets typical camera rates (10–30 Hz).

APPENDIX VII
CALIBRATION PROTOCOL

TABLE VIII: Calibration metrics and baseline hyperparameters.

Item	**Setting**
ECE / ACE / SCE	B=15 bins
ECE binning	15 equal-width bins on $[0, 1]$
ACE binning	15 equal-mass bins (sorted by confidence)
SCE binning	15 equal-width bins, averaged over classes
Temp. Scaling	$T_k \in [0.1, 20]$, per-horizon, NLL objective
Platt Scaling	(a_k, b_k) per-horizon, L-BFGS-B on $\sigma(a_k \mathrm{logit}(p) + b_k)$
Isotonic	`sklearn` IsotonicRegression, $y_{\min}$=0.01, $y_{\max}$=0.99, clip
Hist. Binning	15 equal-width bins
DAC features	CNN penultimate layer (fc2 ReLU), 128-D
DAC kNN	k=10, Euclidean distance
DAC norm.	mean kNN distance, normalized by μ+2σ and clipped to $[0, 1]$
DAC modulation	$T_{\text{eff}} = T_k \cdot \exp(2\, d_n)$